

Spring 2008, Math706 – Numerical Linear Algebra

Homework 5

1. *Gerschgorin's Circle Theorem*: Every eigenvalue of $\mathbf{A} \in \mathbb{C}^{m \times m}$ lies in at least one of the m circular disks in the complex plane with center a_{ii} with radii $\sum_{j \neq i} |a_{ij}|$. Moreover, if n of the disks form a connected domain that is disjoint from the other $m - n$ disks, then there are precisely n eigenvalues of $\mathbf{A}$ within this domain.

- (a) Prove the first part of Gerschgorin's Theorem.
- (b) Prove the second part of Gerschgorin's Theorem.
- (c) Give estimates based on Gerschgorin's Theorem for the eigenvalues of

$$\mathbf{A} = \begin{pmatrix} 8 & 1 & 0 \\ 1 & 4 & \epsilon \\ 0 & \epsilon & 1 \end{pmatrix}, \quad |\epsilon| < 1.$$

- (d) Find a way to establish the tighter bound $|\lambda_3 - 1| \leq \epsilon^2$ on the smallest eigenvalue of $\mathbf{A}$.
2. (a) Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be tridiagonal and hermitian, with all its sub- and superdiagonal entries nonzero. Prove that the eigenvalues of $\mathbf{A}$ are distinct.
- (d) On the other hand, let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be upper-Hessenberg, with all its subdiagonal entries nonzero. Give an example that shows that eigenvalues of $\mathbf{A}$ are not necessarily distinct.
3. Show that for a nonhermitian matrix $\mathbf{A} \in \mathbb{C}^{m \times m}$, the Rayleigh quotient $r(x)$ gives an eigenvalue estimate whose accuracy is generally linear, not quadratic. Explain what convergence rate this suggests for the Rayleigh quotient iteration applied to nonhermitian matrices.
4. A real symmetric matrix $\mathbf{A}$ has an eigenvalue 1 of multiplicity 8, while all the rest of the eigenvalues are ≤ 0.1 in absolute value. Describe an algorithm for finding an orthonormal basis of the 8-dimensional eigenspace corresponding to the dominant eigenvalue.
5. Let $\mathbf{A}, \Delta \mathbf{A} \in \mathbb{R}^{m \times m}$ be symmetric matrices. If $\Delta \mathbf{A}$ is positive definite, show that

$$\lambda_j(\mathbf{A}) \leq \lambda_j(\mathbf{A} + \Delta \mathbf{A}), \quad j = 1, \dots, m$$

where

$$\begin{aligned} \lambda_1(\mathbf{A}) &\geq \lambda_2(\mathbf{A}) \geq \dots \geq \lambda_m(\mathbf{A}), \\ \lambda_1(\mathbf{A} + \Delta \mathbf{A}) &\geq \lambda_2(\mathbf{A} + \Delta \mathbf{A}) \geq \dots \geq \lambda_m(\mathbf{A} + \Delta \mathbf{A}). \end{aligned}$$

6. The problem asks you to put together a MATLAB program that finds all the eigenvalues of a real symmetric matrix.

- (a) Write a function $\mathbf{T} = \mathbf{tridiag}(\mathbf{A})$ that reduces a real symmetric $m \times m$ matrix to tridiagonal form by orthogonal similarity transformations. Your program should use only elementary MATLAB operations. Your output matrix $\mathbf{T}$ should be symmetric and tridiagonal up to rounding errors. Apply your program to $\mathbf{A} = \mathbf{hilb}(5)$.
- (b) Write a function $\mathbf{T}_{\text{new}} = \mathbf{qralg}(\mathbf{T})$ that runs the unshifted QR algorithm on a real tridiagonal matrix $\mathbf{T}$. For the QR factorization at each step, use programs $[\mathbf{W}, \mathbf{R}] = \mathbf{house}(\mathbf{A})$ and $\mathbf{Q} = \mathbf{form}(\mathbf{W})$ from your past homeworks or a new code based on Givens rotations or 2×2 Householder reflections rather than $m \times m$ operations. Your program should stop and return the updated $\mathbf{T}$ when the the (m,m-1)-element satisfies $|t_{m,m-1}| < 10^{-12}$. Apply again your program to $\mathbf{A} = \mathbf{hilb}(5)$.
- (c) Write a driver program which (i) calls $\mathbf{tridiag}$, (ii) calls $\mathbf{qralg}$ to get one eigenvalue, (iii) calls $\mathbf{qralg}$ with a smaller matrix to get another eigenvalue, and so on until all of eigenvalues of $\mathbf{A}$ are determined. The output should be a set of eigenvalues. Run your program for $\mathbf{A} = \mathbf{hilb}(5)$. Print out all eigenvalues and the total number of QR iterations.
- (d) Modify $\mathbf{qralg}$ so that it uses the the Rayleigh quotient shift at each step, then repeat (b) and (c).